A Falsifiable HNC Biomolecule-to-Frequency Packet Hypothesis

for Modulating Oxidative DNA-Damage Markers in Normal and Cancer Cell Models

White Paper / Pre-Experimental Validation Protocol

Framework: HNC BioMolecule -> Adjacent Frequency Packet Blueprint

Working Version: v1.0 white paper compiled from v0.2 biomolecule packet and v0.3 cancer open-data layer

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Claim boundary

This paper reports a computationally reproducible spectral-packet design and a falsifiable in-vitro testing plan. It does not report completed wet-lab results, does not make a clinical claim, and does not claim DNA sequence rebuilding unless future sequencing confirms restoration of a defined allele after treatment.

Abstract

This white paper formalizes the HNC BioMolecule -> Adjacent Frequency Packet framework as a testable hypothesis rather than a therapeutic claim. The framework maps published spectral features of dandelion-associated phenolics - chlorogenic acid, chicoric acid and luteolin - into a reproducible modulation-frequency packet through physical frequency conversion, octave downconversion and adjacent sideband generation. The computational discovery is a 13-tone phenolic coherence packet spanning 1012.1-1817.7 Hz, with densest convergence in the 1367.2-1439.6 Hz region, generated from independent FTIR/Raman/UV-visible peaks. The proposed biological test uses a 660 nm red-light carrier with amplitude modulation and a 10-arm control structure to determine whether the HNC packet adds signal specificity beyond phytochemistry, carrier effects and randomized/scrambled modulation. Primary endpoints are ROS, 8-oxodG, alkaline and enzyme-modified comet assay, gamma-H2AX/53BP1 kinetics, and base-excision-repair markers including OGG1, APE1/APEX1 and XRCC1. The cancer open-data layer selects A549, HCT116 and MCF-7 as priority transformed models, with HeLa reserved as an HPV/p53-rewired context, using DepMap/CCLE/Sanger resources for baseline molecular annotation. The hypothesis is falsified if the full HNC arm does not outperform molecule-only, carrier-only, random-packet and molecule-plus-random-packet controls, or if apparent DNA-damage reduction is explained by cell death, temperature, batch effects or nonspecific photobiomodulation. DNA rebuilding remains a Level 4 claim requiring targeted or single-cell sequencing evidence of restoration and persistence of a defined sequence variant.

Keywords: Taraxacum officinale; chlorogenic acid; chicoric acid; luteolin; spectral conversion; photobiomodulation; oxidative DNA damage; gamma-H2AX; comet assay; DepMap; cancer cell lines; falsifiable hypothesis.

Executive Summary: What Has Been Discovered and What Has Not

Table 1. Discovery status and claim boundary.

Layer	Finding / boundary
Computationally discovered	Published biomolecular spectral peaks can be converted into a reproducible 13-tone HNC phenolic coherence packet. Two dense convergence bands were detected: 1367.2-1402.1 Hz and 1398.6-1439.6 Hz.
Scientifically testable	A 660 nm carrier can be amplitude-modulated by this packet and tested against sham, carrier-only, random-packet and molecule-only controls in normal and cancer cell models.
Not yet discovered	No wet-lab evidence has yet shown that the packet reduces DNA damage, accelerates repair or restores sequence mutations.
Claim ceiling before experiments	The present claim is computational reproducibility and experimental readiness. DNA protection, repair modulation, mutation prevention and DNA rebuilding remain staged hypotheses.

1. Background and Rationale

The originating biological question was whether plant resilience under synthetic signal pressure could be translated into a biomimetic framework for human-cell stress response. The formalized version does not infer human DNA effects directly from plant survival. Instead, it uses plant biomolecules as spectral templates and tests whether a physical carrier encoded with their converted spectral structure modulates measurable oxidative DNA-damage and repair markers.

Dandelion (Taraxacum officinale) is a suitable source plant because open literature describes phenolic acids, flavonoids, polysaccharides, sesquiterpene lactones, sterols and triterpenes across plant organs, with chlorogenic, caffeic, p-coumaric, sinapic, ferulic and chicoric acids reported in HPLC-based studies [1,2]. A 2025 Food & Function study further identified chicoric, caftaric, chlorogenic and caffeic acids in dandelion extracts and reported mitigation of H_2O_2-induced oxidative stress in HypoE22 hypothalamic cells [3].

The strongest physical-carrier bridge is red/near-infrared photobiomodulation rather than UV exposure. PBM literature commonly discusses red/NIR windows around 600-700 nm and 780-1100 nm and proposes mitochondrial, redox and transcriptional mechanisms [7]. A visible red-light study in normal human dermal fibroblasts reported enhancement of GADD45A-mediated DNA repair activity after UV-induced oxidative damage [8]. These papers motivate the carrier but do not validate the HNC conversion algorithm; that validation requires the controls defined below.

2. Conceptual Model

The HNC framework is treated here as a signal-architecture hypothesis. It becomes biologically testable only after conversion into a physical carrier and a measurable cellular endpoint.

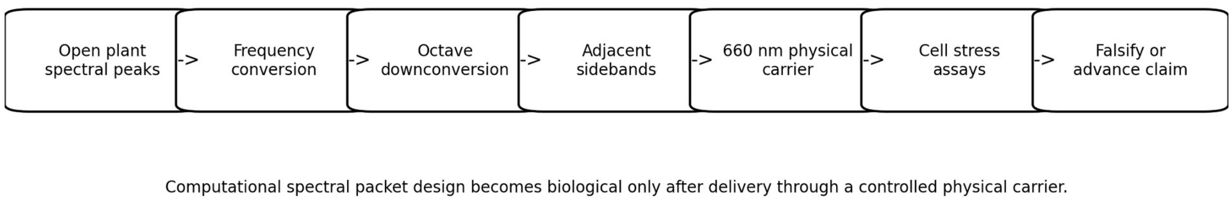

Figure 1. Operational HNC pipeline from open spectral inputs to falsifiable cell-assay outcomes.

Published molecule spectrum -> molecular frequency -> octave-downconverted modulation frequency -> adjacent packet -> physical carrier -> cell stress assay -> statistical falsification or advancement of claim level

3. Spectral Conversion Rules

For FTIR/Raman wavenumbers in cm^-1:

molecular_frequency_THz = wavenumber_cm^-1 * 0.0299792458
infrared_wavelength_um = 10000 / wavenumber_cm^-1

For UV-visible wavelengths in nm:

molecular_frequency_THz = 299792.458 / wavelength_nm

For octave downconversion into an experimentally convenient modulation band:

packet_frequency_Hz = molecular_frequency_Hz / 2^N
Select N so packet_frequency_Hz falls within the pre-specified target band, here 1000-2000 Hz.

Adjacent packets are generated as narrow sidebands and HNC geometric sidebands:

narrow sidebands = f0 * [0.99, 1.00, 1.01]
geometric sidebands = f0 * [1 - phi^-9, 1.00, 1 + phi^-9]
phi = 1.618033988749895; phi^-9 ~= 0.0132

4. Open Spectral Inputs

The v0.2/v0.3 pipeline uses open spectral data. Chlorogenic-acid Raman/IR features at 1603, 1632, 1160 and 1444 cm^-1 are anchored to a 2025 Frontiers in Chemistry Raman/IR study [4]. Chicoric-acid UV-visible conversion uses a maximum near 330 nm and a 300 nm shoulder reported in a Frontiers review [5]. Luteolin spectral assignments use DFT/spectral literature reporting OH, carbonyl and aromatic skeletal vibration regions [6].

Table 2. Conversion table for the HNC phenolic coherence packet.

Molecule	Peak	Assignment	Octaves	Packet Hz
chlorogenic acid	1603 cm^-1	C=C catechol / ethylene vibration	35	1398.6
chlorogenic acid	1632 cm^-1	C=C vibration	35	1423.9
chlorogenic acid	1160 cm^-1	ester C-O deformation	35	1012.1
chlorogenic acid	1444 cm^-1	IR analytical peak	35	1259.9
chicoric acid	330 nm	UV-visible maximum	39	1652.5
chicoric acid	300 nm	UV-visible shoulder	39	1817.7
chicoric acid	1650 cm^-1	carbonyl vibration	35	1439.6
luteolin	1624 cm^-1	carbonyl stretch	35	1417.0
luteolin	3615 cm^-1	OH stretching	36	1577.1
luteolin	1607 cm^-1	aromatic skeletal vibration	35	1402.1
luteolin	1594 cm^-1	aromatic skeletal vibration	35	1390.8
luteolin	1582 cm^-1	aromatic skeletal vibration	35	1380.3
luteolin	1567 cm^-1	aromatic skeletal vibration	35	1367.2

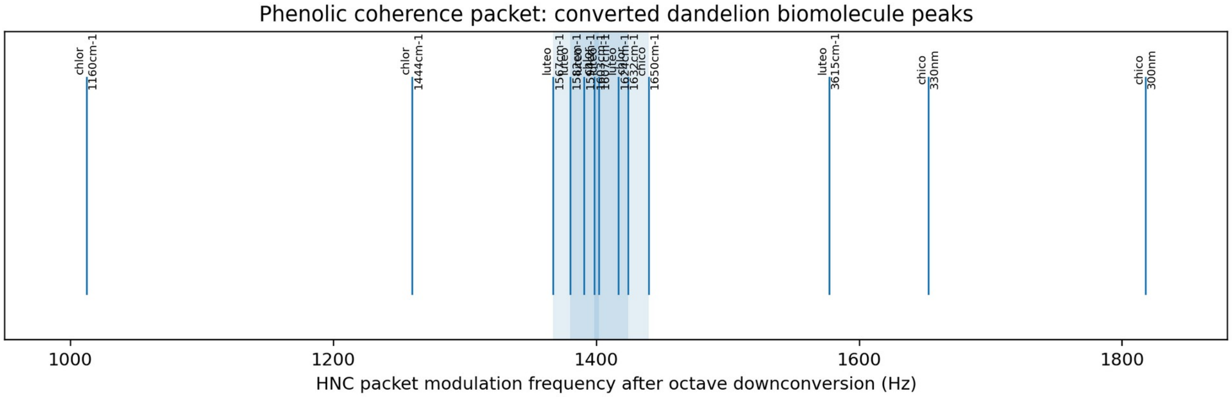

Figure 2. Converted packet frequencies and detected convergence regions.

5. HNC Phenolic Coherence Packet Specification

Table 3. Primary HNC packet specification.

Field	Value
Packet ID	HNC-1E2C1999D7
Plant template	Taraxacum officinale, leaf/root mixed extract
Molecules	chlorogenic acid, chicoric acid, luteolin
Carrier	660 nm red light, amplitude modulated; carrier-only and sham controls required
Core frequencies	1012.1 Hz, 1259.9 Hz, 1367.2 Hz, 1380.3 Hz, 1390.8 Hz, 1398.6 Hz, 1402.1 Hz, 1417.0 Hz, 1423.9 Hz, 1439.6 Hz, 1577.1 Hz, 1652.5 Hz, 1817.7 Hz
Duty cycle	0.6180339887498948
Phase pattern	golden_ratio
Claim ceiling before biological data	Computational packet generation and experimental readiness only

Table 4. Highest-density coherence nodes detected by v0.3.

Coherence range (Hz)	Center (Hz)	Peaks	Molecules
1380.3-1423.9	1402.1	6	chlorogenic acid, luteolin
1367.2-1402.1	1387.8	5	chlorogenic acid, luteolin
1398.6-1439.6	1416.3	5	chicoric acid, chlorogenic acid, luteolin

6. Falsifiable Hypotheses

Table 5. Hypothesis set and claim tiers.

Hypothesis	Falsifiable statement
H0 - null	The HNC packet has no specific biological effect. Arm H is indistinguishable from molecule-only, carrier-only, random-packet and molecule-plus-random-packet controls on pre-specified DNA-damage and repair endpoints.
H1 - DNA protection	In stressed cells, Arm H reduces a composite DNA-damage score more than Arms C, D, E and G while preserving viability and temperature equivalence.

Hypothesis	Falsifiable statement
H2 - repair modulation	Arm H accelerates resolution of gamma-H2AX/53BP1 foci and modulates OGG1, APE1/APEX1, XRCC1, PARP1, GADD45A and NRF2-associated transcripts/proteins relative to controls.
H3 - cancer-context modulation	Effect size and direction differ between normal cells and transformed lines according to baseline NRF2/DNA-repair state. A549, HCT116 and MCF-7 are priority transformed models.
H4 - DNA rebuilding candidate	A Level 4 claim is allowed only if targeted UMI sequencing or single-cell sequencing shows restoration of a defined sequence variant above background and persistence after clonal expansion.

Primary falsification conditions:

- Arm H does not outperform Arm C or Arm G on primary endpoints after correction for multiple comparisons.
- Random or scrambled packets produce the same effect size as the HNC packet.
- The apparent effect is explained by >10-15% excess cell death, thermal drift, irradiance mismatch, batch effects or differential media conditions.
- DNA-damage markers change but sequencing does not show restoration of a defined variant; in that case DNA rebuilding must not be claimed.

7. Experimental Design

Baseline -> Stressor -> Molecule / Carrier / Random packet / HNC packet -> DNA damage and repair endpoints

Decisive test: Arm H (stressor + dandelion molecules + HNC packet) must outperform Arm C, D, E, and G without being explained by cell death, heating, or random/scrambled modulation.

Figure 3. Decisive experimental comparison logic.

Table 6. Ten-arm control matrix.

Arm	Description	Purpose
A	No stressor, no molecule, no signal	baseline
B	Stressor only	damage_control
C	Stressor + standardized dandelion extract	chemical_biomolecule
D	Stressor + 660 nm carrier only	carrier_effect
E	Stressor + random-frequency packet	random_frequency_control
F	Stressor + HNC packet only	signal_only_hnc
G	Stressor + dandelion extract + random packet	molecule_plus_nonspecific_signal
H	Stressor + dandelion extract + HNC packet	full_hypothesis
I	Stressor + purified luteolin/chlorogenic/chicoric acids + HNC packet	molecule_specific_validation
J	Stressor + known assay-positive repair/protection control	assay_validation

Recommended normal-cell models: primary human dermal fibroblasts and/or human keratinocytes. Recommended first transformed models: A549, HCT116 and MCF-7. The first stressor should be H_2O_2 titrated per line to produce robust DNA-damage markers while preserving enough viability to interpret repair kinetics, typically by pre-test dose-response rather than a fixed universal concentration.

8. Cancer Open-Data Layer

Cancer cells are not chosen because reduced DNA damage would be therapeutically favorable. They are chosen because their redox state, checkpoint architecture and DNA-repair wiring are often abnormal, creating a stringent mechanistic test of signal specificity. Public DepMap/CCLE/Sanger resources provide baseline mutation, expression, dependency and lineage annotation for model selection [10-13].

Table 7. Cancer-cell model selection matrix.

Model	DepMap ID	CCLE name	Rationale	Baseline hypothesis
A549	ACH-000681	A549_LUNG	KEAP1-mutant/NRF2-active context tests whether the HNC phenolic packet adds signal over an already redox-adapted cancer state.	NRF2/HO-1/NQO1 may already be high; incremental NRF2 induction may be blunted while ROS and gamma-H2AX kinetics may still shift.
HCT116	ACH-000971	HCT116_LARGE_INTESTINE	Mismatch-repair-deficient but p53-competent colon model; strong literature connection to luteolin and chicoric acid effects.	MMR deficiency and p53 responsiveness may separate DNA-damage marker reduction from apoptosis/cell-cycle effects.
MCF7	ACH-000019	MCF7_BREAST	Common breast cancer model for polyphenol response; useful for comparing redox/DNA damage endpoints against altered apoptosis execution.	Caspase-3 deficiency means apoptosis interpretation requires PARP1, caspase-7, annexin V/PI, mitochondrial and cell-cycle readouts rather than CASP3 alone.
HeLa	ACH-001086	HELA_CERVIX	HPV E6/E7 transformed context tests p53-impaired DNA-damage response under a viral-oncoprotein rewiring regime.	p53-dependent repair/apoptosis signaling may be distorted by HPV oncoprotein activity; gamma-H2AX/comet endpoints are still useful.

Model-specific interpretation controls:

- A549 tests KEAP1/NRF2-high redox adaptation; high baseline NRF2 may blunt further antioxidant-response induction [14].
- HCT116 separates mismatch-repair deficiency from p53-competent DNA-damage/cell-cycle behavior [15].
- MCF-7 requires apoptosis readouts beyond caspase-3 because CASP3 is deficient in this line [16].
- HeLa is a reserve HPV-transformed model in which E6/E7 biology can distort p53/Rb pathway interpretation [17].

9. Primary Endpoints and Assay Readout

Table 8. Endpoint families and interpretation rules.

Endpoint family	Markers / assays	Interpretation role
Redox	ROS/DCFDA, mitochondrial ROS, GSH/GSSG, NFE2L2, KEAP1, HMOX1/HO-1, NQO1	Determines whether packet changes redox stress before DNA markers shift.
DNA damage	8-oxodG, alkaline comet tail moment, hOGG1/Fpg-modified comet, gamma-H2AX and 53BP1 foci	Tracks oxidative base damage, strand breaks and double-strand-break signaling.
DNA repair	OGG1, APE1/APEX1, XRCC1, PARP1, GADD45A, RAD51, BRCA1, TP53, CDKN1A	Determines whether lesion clearance and repair-associated signaling change over time.
Cell fate	Viability, apoptosis/annexin V-PI, PARP cleavage, cell-cycle, senescence	Rules out false marker reduction due to cell death or growth arrest.
Sequence restoration	Targeted UMI sequencing, single-cell sequencing, clonal persistence	Only route to a DNA rebuilding candidate claim.

Base excision repair is the primary pathway for many oxidative base lesions, including 8-oxoguanine; OGG1, APE1, PARP1 and XRCC1 are therefore central readouts [18-20]. Gamma-H2AX/53BP1 foci are used as DNA double-strand-break and repair-kinetic markers [21,22]. OECD AOP 296 links oxidative DNA damage to chromosomal aberrations and mutations and supports oxidative-lesion and comet-style endpoints as regulatory-relevant evidence streams [23].

10. Statistical Analysis Plan

Pre-specify a primary composite DNA-damage score to prevent endpoint shopping:

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Composite_DNA_Damage_Z = mean(z(8-oxodG), z(comet_tail_moment), z(gamma-H2AX_foci), z(53BP1_foci))
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Recommended model:

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endpoint ~ arm * timepoint * cell_line + batch + baseline_damage + viability + temperature + (1 | experiment_day)
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- Primary comparison: Arm H versus C, D, E and G at matched timepoints using planned contrasts with Holm or Benjamini-Hochberg correction.
- Minimum pilot: at least three independent biological experiments; stronger design: n=4-6 independent experiments per line with technical replicates nested within biological replicate.

- Effect-size gate for progression: H vs G standardized mean difference at least 0.3 on the composite score, or a pre-specified 15-20% reduction in key DNA-damage endpoints without excess viability loss.
- Thermal and irradiance controls: plate temperature, optical power density and duty cycle must be logged for every exposure.
- Blinding: waveform file labels should be blinded to the operator and image analyst.

11. Claim-Level Decision Tree

Table 9. Evidence ladder for responsible interpretation.

Claim level	Evidence required	Allowed interpretation
Level 1 - DNA protection	Lower 8-oxodG/comet/gamma-H2AX after stress versus controls	May claim DNA-damage reduction/protection in cell model only.
Level 2 - Repair modulation	Faster marker resolution plus repair-gene/protein kinetics	May claim repair-pathway modulation.
Level 3 - Mutation prevention	Lower micronucleus or mutation frequency after genotoxic stress	May claim mutation-prevention tendency in model.
Level 4 - DNA rebuilding candidate	Sequencing-confirmed restoration of a defined variant and persistence after cell division	May investigate DNA rebuilding mechanism; replication required.

12. Limitations and Risk Controls

- Frequency conversion is a hypothesis-generating encoding, not proof that cells recognize a molecule-derived harmonic signature.
- A 660 nm carrier has its own PBM biology; random and scrambled controls are mandatory to distinguish HNC structure from nonspecific photobiomodulation.
- Polyphenols can act as antioxidants or pro-oxidants depending on concentration, metal context, cell line and stressor dose; dose-response curves are mandatory.
- Cancer-cell DNA-damage reduction may be mechanistically interesting but could also protect malignant cells; no therapeutic conclusion follows from a reduced marker alone.
- DNA rebuilding is not inferred from reduced oxidative markers; it requires direct sequencing evidence.

13. Data and Code Package

The white paper is linked to a reproducible local package generated during framework development:

Table 10. Reproducibility package manifest.

File	Role
hnc_biomolecule_packet_v02.py	Biomolecule packet engine with adaptive octave conversion
hnc_biopacket_phenolic_coherence_v02.json	Primary v0.2 phenolic coherence packet
hnc_conversion_table_v02.csv	Core v0.2 conversion table
hnc_biomolecule_packet_v03_cancer.py	Cancer open-data layer engine
hnc_cancer_packet_spec_v03.json	Full v0.3 cancer packet and model specification
hnc_cancer_conversion_table_v03.csv	Cancer-layer conversion table used in this paper
hnc_cancer_cell_model_matrix_v03.csv	Cell-line selection matrix
hnc_cancer_control_matrix_v03.csv	Ten-arm experimental matrix
hnc_cancer_public_data_hooks_v03.csv	DepMap/CCLE/Sanger/GEO public-data hooks
hnc_cancer_random_frequency_control_v03.json	Random control waveform specification
hnc_cancer_scrambled_phase_control_v03.json	Scrambled phase control waveform specification
hnc_cancer_validation_report_v03.json	Structural validation report

14. Ethics, Biosafety and Non-Clinical Status

The proposed work is in-vitro and pre-clinical. It does not involve human subjects unless primary cells are newly collected, in which case institutional review and consent requirements apply. Cell lines require authentication, mycoplasma testing, passage-number control and biosafety compliance. Light exposure requires optical safety procedures and thermal monitoring. Nothing in this white paper should be interpreted as a recommendation for patient treatment, vaccine alteration, cancer therapy or home experimentation.

15. Conclusion

The HNC BioMolecule -> Adjacent Frequency Packet framework has reached a testable stage. The computational contribution is a reproducible transformation from open spectral biomolecule peaks to a structured 13-tone phenolic coherence packet, plus an open-data cancer-cell model layer and a pre-specified 10-arm falsification design. The next scientific step is not stronger assertion; it is blinded in-vitro testing against random, scrambled, carrier-only and molecule-only controls. The framework advances only if Arm H shows reproducible endpoint-specific effects not explained by nonspecific photobiomodulation, phytochemistry, cell death, heating or batch artifacts. DNA rebuilding remains a separate and higher claim requiring direct sequence restoration evidence.

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Appendix A. Minimal Preregistration Template

- Primary hypothesis: Arm H reduces the composite DNA-damage score more than C, D, E and G under H2O2 stress.
- Primary endpoint: composite DNA-damage z-score at 1, 3, 6 and 24 h.
- Key exclusion: any well with contamination, temperature drift outside pre-specified tolerance or viability below the analysis floor.
- Primary models: one normal line and two transformed lines selected before packet exposure.
- Blinding: waveform identity blinded until image quantification and initial statistics are complete.
- Progression gate: H vs G passes effect-size and adjusted-p-value thresholds in two independent biological replicates before expansion.